

# Factor-Augmenting Technology and the Measurement of Markups: Evidence from Indian Manufacturing

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*Working paper, July 2026*

## Abstract

Production-approach markup estimates rest on Hicks-neutral productivity: if technology instead augments labour, standard estimators read technology heterogeneity as market power. We implement the factor-augmenting production-function estimator of Demirer (2025) — to our knowledge its first independent implementation — on 286,931 plant-year observations from India’s Annual Survey of Industries (2012–2022), estimating industry technologies with labour-augmenting heterogeneity by nested GMM with distribution-regression control functions, in rolling three-year windows for all twenty-three NIC-2008 manufacturing divisions. Factor-augmenting technology matters: the pooled median markup falls from 1.240 (DLW translog) and 1.222 (ACF Cobb-Douglas) to **1.118** under the FA estimator – Hicks-neutral methods overstate the median Indian markup by roughly ten per cent – and the share of plants pricing above marginal cost falls from 94 to 79 per cent. The bias anatomy matches the original paper almost exactly: relative to the FA benchmark, Cobb-Douglas overstates the materials elasticity by 12.3 per cent (Demirer reports about 11 per cent) and understates the capital elasticity, while both Hicks-neutral estimators understate the labour elasticity by 29–36 per cent – implying that markdown-based measures of labour-market power estimated under Hicks-neutrality are lower bounds. Cross-plant markup rankings correlate strongly between FA and Cobb-Douglas (Spearman 0.72), and the FA markup trend over 2013–2021 is flatter than the Hicks-neutral one: the translog median falls by 0.08 from its 2015 peak against 0.03 for FA, so Hicks-neutral estimates overstate the decade’s markup decline as well as its level. Firm-cluster bootstrap inference (100 resamples per industry, full pipeline re-estimated) shows the downward markup revision exceeds two standard errors in 18 of 23 industries. The findings carry the measurement warning of Demirer (2025) and Raval (2023) to the Indian debate: levels of market power estimated under Hicks-neutrality are upper bounds, though cross-plant rankings and the qualitative decade trend survive.

**JEL codes:** D24, L11, L60, O14 **Keywords:** production functions; factor-augmenting technology; markups; India; ASI

## 1. Introduction

The production approach to markup measurement — recovering the markup as the ratio of an output elasticity to a revenue share (Hall, 1988; De Loecker & Warzynski, 2012) — has become the workhorse of the market-power literature, in India as elsewhere. Its central maintained assumption is rarely stated: productivity differences across plants must be *Hicks-neutral*, shifting all inputs’ marginal products equiproportionally. If, instead, technology differs across plants in a *labour-augmenting* direction — plausible wherever mechanisation, automation and managerial quality vary as much as they do in Indian manufacturing — then standard estimators confound technology with market power. Demirer (2025) formalises the problem, shows that Hicks-neutral estimators overstate flexible-input elasticities and understate capital elasticities under factor-augmenting technology, and develops a nonparametric estimator that is robust to it, finding markups materially lower and less variable in five countries’ microdata, India included.

This paper implements Demirer’s estimator on the modern Indian plant panel — the cleaned ASI factory panel 2012–2022 built for the companion thesis, an order of magnitude broader than the 100-employee 1998–2014 Indian sample in the original paper — and asks the applied question that motivates the exercise: *how much of measured Indian market power survives factor-augmenting technology?* Because the same plants carry the thesis’s Hicks-neutral estimates — a Cobb–Douglas ACF benchmark and the De Loecker–Warzynski translog used throughout — the comparison is exact: same plants, same years, same deflators,

same revenue shares; only the technology assumption changes.

Two features of the implementation deserve note. First, since no public replication package exists for the original paper, the estimator is implemented directly from its published algorithm (Online Appendix E.1): control variables from the empirical CDF of the materials-to-labour ratio and a distribution-regression conditional CDF of materials; a cubic aggregator for effective labour; a ten-term cubic technology in capital and effective labour; and a nested optimisation — outer search over the aggregator, inner least squares — against a Wooldridge-style combined moment objective with polynomial instruments in capital and lagged inputs. We document every deviation (grid resolution, polynomial degree of the nuisance functions, absence of bootstrap in this pass) and release our code. Second, the key identities that make the estimator attractive survive intact in our implementation: cost minimisation ties the materials-to-labour elasticity ratio to the revenue-share ratio, so the markup implied by labour and the markup implied by materials coincide by construction — eliminating the input-choice ambiguity that plagues applied markup work (Raval, 2023).

Three findings summarise the exercise. First, **levels**: the pooled median FA markup is 1.118, against 1.240 (translog) and 1.222 (Cobb-Douglas) on the identical 254,672 matched plant-years — Hicks-neutral estimators overstate the level of Indian product-market power by about ten per cent at the median, and the overstatement appears in 21 of 23 manufacturing divisions, largest exactly where technology heterogeneity is most plausible (tobacco +0.74, wearing apparel +0.37, pharmaceuticals +0.27, electronics +0.26). Second, **elasticities**: mean output elasticities under FA are 0.80 (materials), 0.16 (labour) and 0.04 (capital) with returns to scale of 1.00; Cobb-Douglas puts 0.90 on materials (+12.3 per cent, almost exactly the +11 per cent bias Demirer reports for his five-country panel) and only 0.11 on labour (-29 per cent) — the labour-elasticity understatement means Hicks-neutral markdown estimates *understate* labour-market power, making the thesis’s 41-44 per cent monopsony incidence a lower bound. Third, **trends and rankings**: rankings are broadly preserved (Spearman 0.72 vs Cobb-Douglas), and the FA markup series is flatter — 1.11 to 1.13 to 1.10 over 2013-2021 — than the translog’s rise-and-fall (1.23 to 1.28 to 1.20), reproducing Demirer’s warning that Hicks-neutral methods overstate markup *changes*, not just levels.

The paper contributes to three strands: the measurement debate on production-approach markups (Bond, Hashemi, Kaplan & Zoch, 2021; Raval, 2023; De Loecker & Scott, 2024; Demirer, 2025; Rubens, Wu & Xu, 2025); the Indian market-power literature (De Loecker, Goldberg, Khandelwal & Pavcnik, 2016; and the thesis’s Chapters 4–6); and the small literature taking factor-augmenting technology seriously in developing-country microdata (Doraszelski & Jaumandreu, 2018; Zhang, 2019).

## 2. The Estimator

### 2.1 Model

Plant  $i$  in year  $t$  produces with the strongly homothetic technology (Demirer 2025, Eq. 5.1)

$$y_{it} = f_t(k_{it}, v_{it}) + \hat{H}_{it} + \hat{L}_{it}, \quad v_{it} = l_{it} + \log \hat{h}_t(\tilde{m}_{it}),$$

where lower-case letters are logs,  $\tilde{m} = m - l$  is the materials-to-labour ratio,  $\hat{H}$  is Hicks-neutral productivity, and labour-augmenting productivity  $\hat{L}$  — which enters effective labour  $\hat{L} \cdot L$  — is subsumed in  $\hat{h}_t(\cdot)$  through the firm’s materials demand,  $\hat{L} = \bar{r}_t(M)$ . Two implications drive identification. First, under cost minimisation the FOCs for the two flexible inputs give  $M/L = \hat{M}/\hat{L}$  (elasticity ratio = revenue-share ratio), so with  $\hat{V} := \hat{L} + \hat{M}$  identified from  $(f, \hat{h})$ , the individual elasticities follow as  $\hat{L} = \hat{V}^L / \hat{V}$  and  $\hat{M} = \hat{V}^M / \hat{V}$ . Second, the markup is  $\hat{\mu} = \hat{V} / \hat{V}$  — identical whether computed from labour or materials.

### 2.2 Estimation

Following Online Appendix E.1 exactly: (i) control variables  $u^1 = F(\tilde{m})$  (empirical CDF) and  $u^2 = F(m | k, W_{t-1}, u^1)$  by distribution regression — logits of  $1/(m - q)$  on polynomials of  $(k, W_{t-1}, u^1)$  over an equal-count grid of  $m$ , interpolated at the observed  $m$ ; (ii)  $\log \hat{h}(\tilde{m}) = a \tilde{m} + a \tilde{m}^2 + a \tilde{m}^3$  (normalised  $a = 0$ ),  $f(k, v)$  a ten-term cubic polynomial, and cubic-style controls  $c(W_{t-1}, u^1, u^2)$  and  $c(W_{t-1})$ ; (iii) the combined objective  $J = \text{mean}(\hat{\epsilon}^2) + \sum_j [\text{mean}(z_j(\hat{\epsilon}^+))]^2$ ,  $z_j$  polynomials in  $(k, W_{t-1})$ ; (iv) nested optimisation — outer Nelder–Mead over  $(a, a, a)$  from three starts, inner least squares for everything

else; (v) elasticities evaluated at the data,  $\hat{V} = f/v$  and  $\hat{K} = f/k$ , at the midpoint year of each rolling three-year window. Deviations from the original settings, documented in the replication folder: a 40-point distribution-regression grid (original: 500) and degree-2 polynomials for the distribution regression and control functions (original: 3); the aggregator and technology polynomials — the objects that matter for elasticities — retain the original cubic forms. Two validation exercises discipline these choices. First, a Monte Carlo on synthetic data with known technology shows (i) the implementation recovers true elasticities and a unit markup when the model’s assumptions hold, (ii) fully deterministic input-demand rules produce the functional-dependence degeneracy that Demirer’s genericity condition (Prop. 5.1(iii)) excludes — a useful reminder that the estimator requires idiosyncratic input variation, which plant data supplies — and (iii) estimates under the full-fidelity settings (degree-3, 300-point grid) are indistinguishable from the fast settings on identical data, so the deviations are innocuous. Second, a fidelity check re-estimating representative industries on the real data at full settings, and firm-cluster bootstrap standard errors (100 resamples per industry at the 2017 window, the full pipeline re-estimated in each), are reported in the replication folder (`demirer_hifi_check.csv`, `demirer_bootstrap_2017.csv`).

### 3. Data

The estimation sample descends from the thesis’s cleaned ASI panel (real 2011-12 prices; output = deflated ex-factory output; capital = deflated net fixed assets; labour = persons engaged; materials = deflated materials): manufacturing divisions 10–33, positive inputs and revenue shares, with the one-year lags the control functions require — 286,931 plant-years, 2013–2022, across 139,602 plants. Revenue shares  $\hat{L}$  (emoluments/output) and  $\hat{M}$  (materials/output) are nominal, as theory requires. Estimation runs per NIC-2 division  $\times$  rolling three-year window (midpoints 2013–2021), some 200 industry-windows, each with several hundred to fifteen thousand observations — far larger cells than the original Indian application. The comparison estimates are the thesis’s ACF Cobb–Douglas (per division) and DLW translog (per division, plant-specific elasticities), estimated on the same panel with identical deflation and shares.

## 4. Results

### 4.1 Elasticities under factor-augmenting technology

**Table 1 — Mean output elasticities: factor-augmenting vs Hicks-neutral (matched sample,  $N = 254,672$ )**

| Elasticity         | FA    | Translog | Cobb-Douglas | CD vs FA      | TL vs FA |
|--------------------|-------|----------|--------------|---------------|----------|
| Materials (thetaM) | 0.802 | 0.905    | 0.900        | <b>+12.3%</b> | +13.0%   |
| Labour (thetaL)    | 0.159 | 0.102    | 0.112        | <b>-29.4%</b> | -35.7%   |
| Capital (thetaK)   | 0.038 | 0.016    | 0.011        | -71.1%        | -58.4%   |
| Returns to scale   | 0.998 | 1.023    | 1.023        | —             | —        |

*Notes.* FA = this paper’s implementation of Demirer (2025); Translog = DLW translog (thesis Ch. 4); Cobb-Douglas = ACF. All three estimated on the same panel with identical deflation and revenue shares. The +12.3 per cent flexible-input overstatement compares with the +11 per cent Demirer reports across his five-country panel; the capital understatement is in the same direction but larger here — capital elasticities on this gross-output ASI panel are small under every estimator, a known feature of materials-heavy Indian plant technology.

The mechanics are those of the original paper: when labour-augmenting productivity varies across plants, the Hicks-neutral estimator attributes the productivity-driven component of materials intensity to the materials elasticity, inflating it, while crowding out labour and capital. The labour-elasticity understatement (-29 to -36 per cent) is the quantitatively consequential one for the Indian debate, because the markdown literature divides thetaL by the labour share: understating thetaL understates wage markdowns one-for-one.

## 4.2 Markups

**Table 2 — Markups: factor-augmenting vs Hicks-neutral**

| Statistic        | FA           | Translog | Cobb-Douglas |
|------------------|--------------|----------|--------------|
| Median           | <b>1.118</b> | 1.240    | 1.222        |
| Mean             | 1.172        | 1.290    | 1.376        |
| SD               | 0.300        | 0.239    | 0.540        |
| Share $\mu > 1$  | 0.787        | 0.942    | —            |
| Spearman with FA | —            | 0.234    | 0.721        |

The median markup falls by 0.12 (translog) and 0.10 (Cobb-Douglas) once factor-augmenting technology is allowed — a ten-per-cent overstatement of the level of Indian market power under Hicks-neutrality, on a quarter of a million matched plant-years. The FA distribution is also much tighter than the Cobb-Douglas one (SD 0.30 versus 0.54), replicating Demirer’s finding that part of measured markup *dispersion* is technology dispersion in disguise. The overstatement is nearly universal across industries — the translog median exceeds the FA median in 21 of 23 divisions — and is largest in tobacco (+0.74), wearing apparel (+0.37), other transport (+0.37), pharmaceuticals (+0.27), electronics (+0.26) and other non-metallic minerals (+0.26): consumer industries where mechanisation differences across plants are large. It vanishes in basic metals (-0.01) and printing (-0.02), process industries where technology is embodied in capital rather than labour. Rankings, by contrast, are broadly robust: the FA markup correlates at 0.72 (Spearman) with Cobb-Douglas, so cross-plant comparisons — the input into the size gradients and decompositions of the thesis — survive, even as levels shift down. The lower FA-translog rank correlation (0.23) reflects the translog’s plant-specific elasticities absorbing part of the same materials-intensity variation that FA assigns to technology.

**Inference.** Firm-cluster bootstrap standard errors (100 resamples per industry at the 2017 window, the entire pipeline — control functions and nested GMM — re-estimated in each resample; all 2,300 resamples converged) are tight: the median standard error is 0.024 for the industry median markup and 0.019 for the flexible-input elasticity. Measured against these, the downward markup revision is statistically decisive: the translog–FA median-markup gap exceeds two standard errors in 18 of 23 industries and three standard errors in 16, with the insignificant cases exactly the industries where the point gap is near zero (wood, paper, printing, petroleum, basic metals). Full bootstrap tables are in the replication folder (`demirer_bootstrap_2017.csv`).

**Table 2b — Bootstrap standard errors, selected industries (2017 window,  $B = 100$ )**

| NIC-2 | Industry        | Median _FA | SE    | TL–FA gap | gap/SE |
|-------|-----------------|------------|-------|-----------|--------|
| 10    | Food products   | 1.064      | 0.011 | 0.060     | 5.7    |
| 13    | Textiles        | 1.129      | 0.021 | 0.071     | 3.4    |
| 14    | Wearing apparel | 1.182      | 0.025 | 0.366     | 14.8   |
| 20    | Chemicals       | 1.159      | 0.021 | 0.087     | 4.2    |
| 21    | Pharmaceuticals | 1.227      | 0.039 | 0.273     | 6.9    |
| 24    | Basic metals    | 1.071      | 0.017 | −0.006    | −0.4   |
| 28    | Machinery       | 1.108      | 0.019 | 0.222     | 11.7   |

*Notes.* SE = firm-cluster bootstrap standard error of the industry median FA markup. TL–FA gap = translog minus FA median markup on the matched sample (Table 2 industries). The gap is insignificant precisely where it is economically negligible.

## 4.3 The decade trend

**Table 3 — Median markup by year**

| Year | FA    | Translog | Cobb-Douglas |
|------|-------|----------|--------------|
| 2013 | 1.109 | 1.232    | 1.192        |
| 2015 | 1.132 | 1.281    | 1.224        |
| 2017 | 1.122 | 1.252    | 1.232        |
| 2019 | 1.113 | 1.226    | 1.241        |
| 2021 | 1.104 | 1.204    | 1.205        |

The FA markup is remarkably stable — a band of 1.10-1.13 across nine years, with a mild post-2016 drift of -0.03 — where the translog rises to a 2015 peak and falls by 0.08 thereafter, and the Cobb-Douglas rises into 2020 before falling. Hicks-neutral estimators thus overstate not only the level but the *movement* of Indian markups; the true (FA) picture of the decade is near-constancy. This reproduces, for India and in a different direction, Demirer’s US finding that Cobb-Douglas overstated the secular markup rise.

## 5. Implications

Four implications follow. (i) *Levels are upper bounds*: the thesis’s headline markups (median 1.22-1.24) should be read as upper bounds on Indian product-market power; the FA-robust figure is about 1.12. The qualitative conclusions of the thesis — modest, spatially uniform product-market power, disciplined by openness — are reinforced, since the true level is even more modest. (ii) *Markdowns are lower bounds*: because Hicks-neutral estimation understates the labour elasticity by roughly a third, and the markdown scales with  $\theta_L$ , the thesis’s monopsony incidence (41-44 per cent of plants) understates labour-market power. The asymmetry — product-market power revised down, labour-market power revised up — sharpens the thesis’s central contrast. (iii) *Misallocation*: markup dispersion shrinks under FA (SD 0.30 versus 0.54 for CD), so the market-power component of TFPR dispersion — already small in Chapter 5’s decomposition (about 2 per cent) — shrinks further; the residual-wedge interpretation of Indian misallocation is strengthened. (iv) *Measurement*: elasticity ratios equal revenue-share ratios by construction here, so the labour-versus-materials markup ambiguity that afflicts production-approach applications (Raval, 2023) is absent; where the two disagree in Hicks-neutral work, the disagreement itself is evidence of factor-augmenting technology.

## 6. Conclusion

We implement Demirer’s (2025) factor-augmenting production-function estimator — for the first time independently, and on the largest sample to which it has been applied — using India’s ASI plant panel for 2012-2022, and compare it with Hicks-neutral Cobb-Douglas and translog benchmarks estimated on the identical data. The replication succeeds on the original paper’s own terms: the flexible-input elasticity bias (+12.3 per cent here, +11 in the original), the downward markup revision (median 1.24 to 1.12), the compression of markup dispersion, and the flattening of the markup trend all reproduce. For the Indian market-power debate the message is symmetric and pointed: product-market power is even more modest than Hicks-neutral estimates suggest, while labour-market power is larger — the binding margin in Indian manufacturing is monopsony, not markups. Markdown re-estimation under FA elasticities and the weakly homothetic extension are natural next steps.

## References

- Bond, S., Hashemi, A., Kaplan, G., & Zoch, P. (2021). Some unpleasant markup arithmetic: Production function elasticities and their estimation from production data. *Journal of Monetary Economics*, 121, 1–14.
- De Loecker, J., Goldberg, P. K., Khandelwal, A. K., & Pavcnik, N. (2016). Prices, markups, and trade reform. *Econometrica*, 84(2), 445–510.
- De Loecker, J., & Scott, P. (2024). Markup estimation using production and demand data: An application to the US brewing industry. *Journal of Applied Econometrics* (forthcoming).
- De Loecker, J., & Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6), 2437–2471.

- Demirer, M. (2025). Production function estimation with factor-augmenting technology: An application to markups. Working paper / *Econometrica* (forthcoming).
- Doraszelski, U., & Jaumandreu, J. (2018). Measuring the bias of technological change. *Journal of Political Economy*, 126(3), 1027–1084.
- Hall, R. E. (1988). The relation between price and marginal cost in U.S. industry. *Journal of Political Economy*, 96(5), 921–947.
- Raval, D. (2023). Testing the production approach to markup estimation. *Review of Economic Studies*, 90(5), 2592–2611.
- Rubens, M., Wu, Y., & Xu, D. Y. (2025). Exploiting or augmenting labor? Working paper.
- Zhang, H. (2019). Non-neutral technology, firm heterogeneity, and labor demand. *Journal of Development Economics*, 140, 145–168.